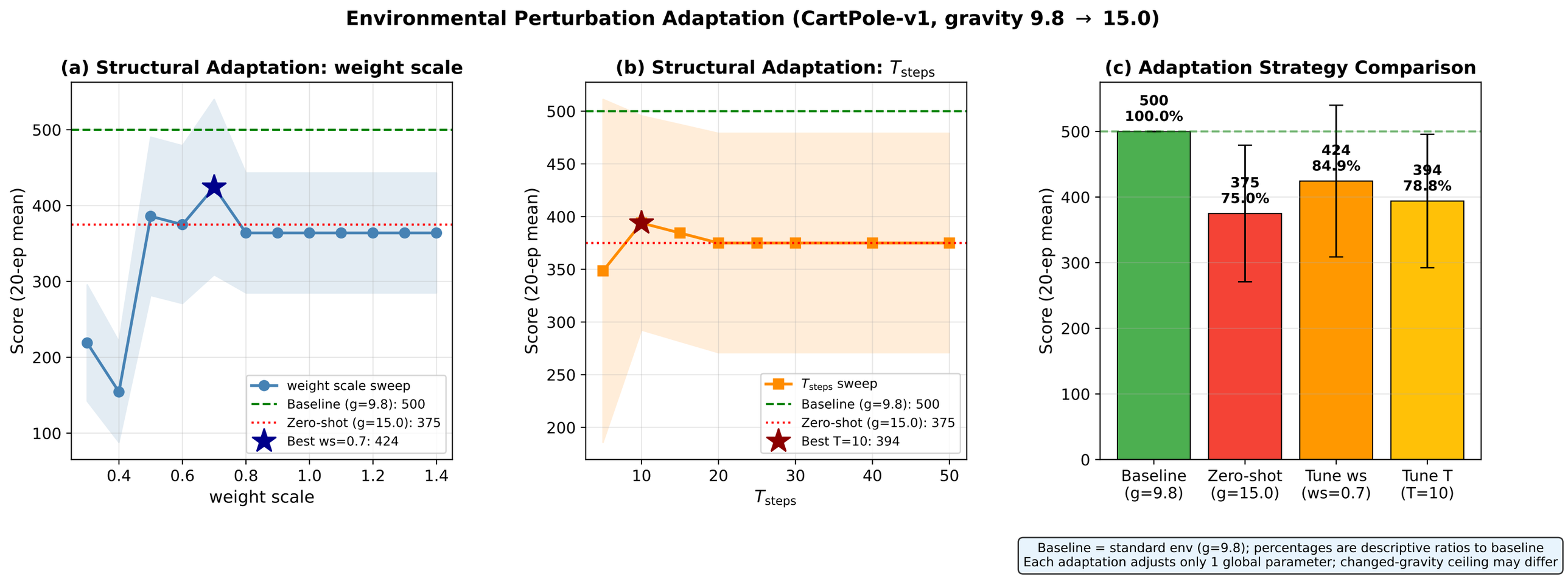
Environmental Perturbation Adaptation (CartPole-v1, gravity 9.8 → 15.0)
(a) Structural Adaptation: weight scale
Score (20-ep mean)
weight scale
weight scale sweep
Baseline (g=9.8): 500
Zero-shot (g=15.0): 375
Best ws=0.7: 424
(b) Structural Adaptation: T_{steps}
T_{steps}
T_{steps} sweep
Best T=10: 394
(c) Adaptation Strategy Comparison
500
100.0%
375
75.0%
424
84.9%
394
78.8%
Baseline
(g=9.8)
Zero-shot
(g=15.0)
Tune ws
(ws=0.7)
Tune T
(T=10)
Baseline = standard env (g=9.8); percentages are descriptive ratios to baseline
Each adaptation adjusts only 1 global parameter; changed-gravity ceiling may differ